

Quantitative Portfolio Management Techniques

INTRODUCTION

Transaction cost analysis (TCA) has become an important decision-making tool for portfolio managers. It allows managers to uncover hidden opportunities that may otherwise not be as transparent, especially given the vast array of data propagating the marketplace. Portfolio managers who once treated transaction costs as an unavoidable cost of business have turned to TCA as a valuable source of incremental alpha. TCA has finally made it to mainstream portfolio management.

Below are just a few ways that TCA is being incorporated into the stock selection phase of the investment cycle.

Quantitative Overlays. Managers select the universe of stocks for potential inclusion into the portfolio. They then determine a subset of stocks from that universe that meet specified investment criteria such as market cap, price to earnings, book value, forecasted profit, etc. As a final filter, managers further reduce the potential investment list based on the expected trading cost. Stocks that are too expensive to transact are eliminated from potential inclusion into the portfolio.

MI Factor Scores. Incorporates both liquidity and volatility to determine a market impact factor score to rank stocks based on trading cost. The higher the score the more expensive it is to transact the stock. The MI factor score provides an equal and fair comparison across all stocks. We show below that MI factor scores provide a large benefit over other techniques that simply rely solely on liquidity or volatility (such as a maximum % ADV to hold in the portfolio).

Cost Curves. Cost curves provide managers with the expected market impact cost for various share quantities and execution strategies. Share quantities are usually expressed in terms of percentage of ADV and the